Veer Requirements Document

Course: Computer Science Senior Design I
Advisor: Dr. Jillian Aurisano
Team: Grady Algire, Brady Cooper, Briar Elliot, Dominic Rowland, Aiden Ward
Document version: 1.0 | October 30, 2026

1. Version and Change Log

Version	Date	Change	Author
0.1	2026-09-26	User stories US-01 to US-05, use cases UC-01 and UC-02, acceptance criteria AC-01.1 to AC-02.2 (User_Stories.md).	Team
0.2	2026-10-02	D0: explicit spoken "okay" required; silence no longer approves a route (decision D07). Supersedes UC-01 step 6.	Team
0.3	2026-10-13	D2: non-functional requirements NFR-01 to NFR-09 and MoSCoW priorities for stories and acceptance criteria (D2 §4).	Team
1.0	2026-10-30	Consolidated into this document. Functional requirements given stable REQ-## IDs; acceptance criteria restated as requirements; NFR-01 to NFR-09 kept with their D2 IDs and given thresholds; NFR-10 (performance) and NFR-11 (speech accuracy) added; constraints given CON-## IDs. Priorities rebalanced so Must is half of the set: REQ-02, REQ-04, REQ-05, REQ-14, and NFR-09 moved from Must (D2) to Should, because each is either covered by another Must or not needed for a working first demo. Usage retention of 90 days / 12 months approved (closes G-05).	Aiden Ward

2. Purpose and Scope

Purpose. This document states what Veer must do and for whom. It is the single list of requirements that the Technical Specification (Technical_Specification.md) traces to and that the Week 10 test plan verifies. It adds no new design.

Product. Veer is an Android navigation app for a driver using a phone mounted in their own car. The driver speaks requests such as "I'm hungry for McDonald's" or "avoid the rain." Veer turns each request into a typed routing constraint, proposes a revised route, reads back a one-sentence summary, and switches routes only after the driver says "okay." Raw audio is never stored, and a passenger can mute the microphone with one tap.

In scope for the first release (Project_Description.md, D0 §1):

- Voice capture with a visible listening indicator and mute
- Transcription and constraint extraction from natural speech
- Waypoint insertion by business or category
- Route preferences: avoid highways, avoid tolls, avoid rain
- Turn-by-turn routing with spoken guidance
- Per-trip AI, Maps, and weather usage tracking for the development team
- A mobile app on the driver's existing Android phone

Out of scope: see the Won't list in Section 7.

3. Stakeholders and User Goals

From User_Stories.md (Week 4).

Category	Stakeholder	Goal	Story
Primary	Driver	Add a food stop by saying what they want, without leaving the route by more than a few minutes.	US-01
Primary	Driver	Tell Veer to avoid rain and spend less time driving in bad weather.	US-02
Secondary	Veer development team	See how many AI and Maps calls each trip uses, to stay within budget.	US-03
Secondary	Google Maps Platform	Routes and places used within its terms (caching, base map).	CON-04
Hidden	Passenger	Know when the app is listening and be able to stop it.	US-04
Hidden	Driver with an accent or speech difference	Have Veer confirm what it heard before changing the route.	US-05
Hidden	Ohio law (ORC 4511.204)	No handheld phone use while driving.	CON-03
Hidden	Businesses shown as stops	Selected only by stated preference and detour cost.	CON-06

4. Glossary

Each term is defined once here and used with this meaning in both documents.

Term	Definition
Active route	The route the driver is following now. Only C1 can replace it.
Allow-list	The fixed set of constraint types and intents C2 accepts from the AI provider. Anything else is rejected.
Approval	The driver saying "okay" after a route proposal. Silence or an unclear reply is not approval.
C1 to C4	Components the team builds: C1 Mobile Navigation, C2 Speech Understanding, C3 Route Planning, C4 Usage Monitor.
Constraint	A typed routing rule from speech: `add_waypoint`, `avoid_precipitation`, `avoid_highways`, or `avoid_tolls`.
Detour (extra time)	Added travel time, in seconds, of a proposed route compared with the active route.
Developer	A member of the Veer development team using the usage report.
Driver	The person driving with Veer running on a phone mounted in their own car. The primary user.
E1 to E4	External dependencies: E1 Phone Platform (Android), E2 AI Provider (OpenAI), E3 Google Maps Platform, E4 Weather Provider (Open-Meteo).
Edge case (EC-##)	A boundary, timing, bad-input, or dependency failure case listed in D2 §3.1.
I1 to I9	Interfaces between components, defined in D0 §4 and revised in D2 (I1 split into I1a, I1b, I1c).
Installation	One anonymous app install, identified by a random installId. Veer has no driver accounts.
Listening indicator	The on-screen and notification display of the mic state: listening, muted, or error.
Monthly cap	The team-set spending limit per service (AI, Maps, weather) per calendar month.
Mute	Stopping the microphone by one tap or the phrase "Veer, stop listening."
MuteError	The state entered when the mic does not stop after a mute request.
Passenger	A person in the car who is not a user of Veer and has not agreed to anything.
Raw audio	Microphone samples. Never written to disk, logs, or any server store.
Recovery	The C1 state that speaks a short prompt and keeps the active route after any failure.
Route proposal	A candidate route from C3, with optional place, extra time, confirmation text, and warnings. Not active until approved.
Trip	One drive from start to arrival or end. Constraints and route state last only for the trip.
Trip revision	A counter that goes up by 1 each time a route is activated. Replies with an older revision are ignored.
Usage event	A record of one AI, Maps, or weather call attempt: service, operation, units, estimated cost, outcome.
Utterance	One spoken request, capped at 15 seconds of audio.

5. Functional Requirements

Verification methods: **T** = test, **D** = demonstration, **I** = inspection, **A** = analysis. "Source" gives the story and, where one exists, the acceptance criterion or design decision the requirement restates.

ID	Requirement	Priority	Source	Verification
REQ-01	The system shall, when the driver asks by voice for a business or category of stop, propose a route that adds a matching place with a detour of 600 seconds (10 minutes) or less.	Must	US-01; AC-01.1	T
REQ-02	The system shall speak a confirmation of 15 words or fewer for every route proposal before it can be activated.	Should	US-05; AC-01.1; EC-05	T
REQ-03	The system shall activate a proposed route only after the driver says "okay"; silence or an unclear reply shall leave the active route unchanged in 100% of trials.	Must	US-05; D07	T
REQ-04	The system shall make an approved route active in both the map and turn-by-turn guidance within 3 seconds after the driver finishes saying "okay."	Should	US-01; AC-01.1	T
REQ-05	The system shall, when no matching place adds 600 seconds or less, keep the active route and ask for a different place within 3 seconds of the request.	Should	US-01; AC-01.3	T
REQ-06	The system shall, when speech cannot be turned into an allow-listed constraint, keep the active route and ask the driver to repeat within 2 seconds after speech ends.	Must	US-05; AC-01.4	T
REQ-07	The system shall, when the driver says "no" to a proposal, offer the next-best candidate in one sentence.	Should	US-01 (UC-01 alternate flow)	T
REQ-08	The system shall, when the driver asks to avoid rain, rank candidate routes by lower precipitation exposure within the configured weather detour limit, and shall warn rather than claim avoidance when weather data is unavailable or stale.	Should	US-02	T (replayed weather data)
REQ-09	The system shall apply spoken "avoid highways" and "avoid tolls" requests by ranking routes that satisfy the flag ahead of routes that do not.	Could	Project_Description.md first-release scope (no story yet; see Q-02)	T
REQ-10	The system shall show the mic state (listening, muted, or error) on screen and in the trip notification for 100% of the time a trip is active.	Must	US-04	I, D
REQ-11	The system shall stop the microphone within 1 second of a mute tap or a locally recognized "Veer, stop listening," and then show muted.	Must	US-04; AC-02.1	T

REQ-12	The system shall, if the microphone has not stopped, force-stop it within 2 seconds of the original mute request, show an error, and send or store 0 seconds of audio captured after the request.	Must	US-04; AC-02.2	T
REQ-13	The system shall restart listening only after an unmute tap.	Should	US-04 (UC-02 alternate flow)	D
REQ-14	The system shall record every AI, Maps, and weather call attempt (including retries and failures) against its trip, and return per-trip call counts and estimated cost to an authenticated developer through the usage report.	Should	US-03	T
REQ-15	The system shall provide turn-by-turn guidance with spoken directions from the current location to a destination chosen before driving.	Must	US-01 and US-02 (precondition: an active route); Project_Description.md scope	D

6. Non-Functional Requirements

NFR-01 to NFR-09 keep the IDs and priorities the team set in D2 §4.3; each has been restated with a measurable condition. NFR-10 and NFR-11 are new in this version.

ID	Type	Requirement	Priority	Source	Verification
NFR-01	Security (privacy)	The system shall write 0 raw audio files to the phone, backend, logs, or database, and shall use E2 only under a no-retention setting confirmed in writing.	Must	US-04; AC-01.2	T (storage scan after trip), A (provider terms)
NFR-02	Reliability	The system shall keep the active route and continue guidance after any failed interpretation, planning, authentication, or provider call, in 100% of edge cases EC-01 to EC-34.	Must	US-01, US-02, US-05	T
NFR-03	Security	The app shall hold no provider API keys; it shall store only a refresh token, in the Android Keystore, and access tokens shall expire after 1 hour. Registration shall be rate-limited per IP address.	Must	US-03	I, T
NFR-04	Reliability (cost)	The system shall alert developers once at 80% and deny paid calls at 100% of each service's monthly cap, and concurrent calls shall never push spending past the cap.	Must	US-03	T
NFR-05	Security (logging)	Every server log line shall be JSON with a requestId, and 0 log lines shall contain audio, transcripts, raw GPS coordinates, tokens, or API keys.	Should	US-03	T (redaction test)
NFR-06	Maintainability	All timeouts, weather thresholds, spending caps, and retention periods shall be configuration values, and the backend shall refuse to start if any value is missing or out of range.	Should	US-02, US-03	I, T
NFR-07	Usability (distraction)	Every spoken reply shall be one sentence of 15 words or fewer, and no feature shall require the driver to touch the screen while the car is moving.	Must	US-01; CON-03; CON-05	I, T
NFR-08	Security (retention)	The system shall delete usage events after 90 days and per-trip totals after 12 months.	Should	US-03	T
NFR-09	Reliability	The microphone and network shall keep working when the app is in the background during a trip, with the foreground trip notification visible for the whole trip.	Should	US-04	T
NFR-10	Performance	C2 shall return an I2 result within 1.5 seconds of utterance end, and C3 shall return an I4 result within 2.5 seconds of the request, for 95% of requests with 5 concurrent trips. These budgets leave room inside the 2-second and 3-second prompt deadlines in REQ-05 and REQ-06.	Should	US-01, US-05	T (load test)
NFR-11	Usability (accuracy)	The system shall turn at least 8 of 10 scripted test phrases into the correct constraint, including phrases from accented speakers.	Should	US-05	T

7. MoSCoW Summary

12 of 26 requirements (46%) are Must. The Must set is the smallest version of Veer that is safe to demo in a car: add a stop by voice, never change the route without "okay," keep the route on failure, show and honor mute, keep no audio, keep provider keys off the phone, and stay under budget.

Must: REQ-01, REQ-03, REQ-06, REQ-10, REQ-11, REQ-12, REQ-15, NFR-01, NFR-02, NFR-03, NFR-04, NFR-07

Should: REQ-02, REQ-04, REQ-05, REQ-07, REQ-08, REQ-13, REQ-14, NFR-05, NFR-06, NFR-08, NFR-09, NFR-10, NFR-11

Could: REQ-09

Won't (this release). Each item is a decision the team already made; it stays out unless this list is changed and logged.

- Driver accounts, saved trips, or history across devices (D0 §1; D2 R-05)
- A web trip planner (D0 §1)
- An iOS app (D1 §6.1)
- Offline map or routing (D0 D08)

- Sponsored or popularity-based ranking of businesses (D0 D10; CON-06)
- Custom in-dash or vehicle hardware (CON-01)
- Storing raw audio or trip conversations for debugging (D0 D03)
- Storing weather history (D1 §2)
- Any mute path that depends on the cloud (D0 D02)
- Microservice deployment of C2 to C4 (D0 §6)

8. Inherited Constraints

From Constraints_Essay.md (Week 3). The Technical Specification §5 states what each one rules out or forces.

ID	Type	Constraint
CON-01	Economic	Veer is funded by the team. It uses free-tier hosted inference and Google Maps Platform credits, and runs on the driver's existing phone rather than purpose-built hardware.
CON-02	Security	Raw audio is not retained: each utterance is transcribed, turned into a constraint, and discarded, and only the constraint is kept for the trip. A visible listening indicator and one-tap mute are required before any in-car demo.
CON-03	Legal	Ohio ORC 4511.204 makes handling a phone while driving a primary offense but permits voice-operated use. No feature may ask the driver to tap the screen while moving.
CON-04	Legal	Google Maps Platform terms bar caching most place data beyond 30 days and showing its results over a competing base map.
CON-05	Ethical	Every response competes with the road for attention, so replies are one short sentence; longer content goes to the screen for a passenger or a stop.
CON-06	Ethical	Businesses are ranked only by the driver's stated preference and detour cost.

9. Assumptions and Open Questions

Assumptions (from UC-01 preconditions and D0/D1):

- A1. The driver chooses a destination before driving (D0 §4 notes).
- A2. The phone has a data connection for interpretation and replanning; guidance continues on the stored route without one (EC-33).
- A3. The phone runs Android and is mounted where the mic can hear the driver (D1 §6.1).
- A4. Hosted inference meets the no-retention requirement; if not, transcription moves to the phone (RK-01 fallback).

Open questions. Owners and due dates follow D2 §1.2 and the Week 8 plan.

ID	Question	Affects	Owner	Due
G-01	Can Android recognize "Veer, stop listening" on the phone within 1 second? Fallback: tap-only mute.	REQ-11	Dominic Rowland (T18)	2026-11-11
G-02	Rain threshold, weather freshness window, and maximum weather detour.	REQ-08, NFR-06	Dominic Rowland (T16)	2026-11-23
G-03	Written proof of the OpenAI no-retention setting for audio.	NFR-01	Briar Elliot (T6)	2026-11-11
G-04	Do Google Maps terms allow turn-by-turn guidance in our own app? Fallback: hand off to Google Maps.	REQ-15, CON-04	Brady Cooper (T11)	2026-11-11
G-05	Usage retention. Closed in v1.0: 90 days for events, 12 months for trip totals.	NFR-08	Aiden Ward	Closed 2026-10-30
G-06	Does a foreground service keep the mic and network alive in the background?	NFR-09	Dominic Rowland (T18)	2026-11-11
G-07	Monthly spending cap amounts for AI, Maps, and weather.	NFR-04	Aiden Ward (T25)	2027-02-25
G-08	Audio encoding and wire format for I2/I3.	NFR-10	Briar Elliot (T6)	2026-11-11
Q-01	Must set rebalanced in v1.0 (Section 7). Revisit if a Should becomes blocking.	All	Team	2027-01-11 (start of build)
Q-02	Add a user story for avoiding highways and tolls (REQ-09), which is in the project scope but has no story.	REQ-09	Brady Cooper	2026-11-11
Q-03	Check the NFR-10 latency targets against first OpenAI and Google Maps measurements.	NFR-10	Briar Elliot, Brady Cooper	2027-02-12 (M-18)
Q-04	User_Stories.md UC-01 step 6 still lets silence approve a route, which D07 superseded. Update the use case.	REQ-03	Aiden Ward	2026-11-11
Q-05	What happens when C4's 500-event queue fills during a database outage (D2 §3.4)?	NFR-04	Aiden Ward	2027-02-15 (T22)